Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/30/2005
PI Name:	Glotzer, Sharon C Ph.D.		
Project Title:	ROLE OF SPATIALLY HETEROGENEOUS DYNAMICS AND POLYDISPERSITY IN THE EARLY STAGES OF HOMOGENEOUS NUCLEATION: APPLICATION TO THE PROMOTION AND SUPPRESSION OF CRYSTALLIZATION AND VITRIFICATION		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sglotzer@umich.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	734-615-6296
Organization Name:	University of Michigan		
PI Address 1:	Mail Stop 3074 H.H. Dow		
PI Address 2:	2300 Hayward St.		
PI Web Page:			
City:	Ann Arbor	**State:**	MI
Zip Code:	48109-2136	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rogers, Richard	**Contact Phone:**	216-433-6512
Contact Email:	Richard.B.Rogers@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Solomon, Michael J. (University of Michigan)		
Grant/Contract No.:	NNC04GA43G		
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Task Description:	This task carries out ground-breaking research on the solidification behavior of supercooled liquids and colloids above their glass transition using molecular dynamics computer simulations and confocal laser scanning microscopy experiments. Homogeneous nucleation phenomena and its relation to local structural and dynamical features is studied. The suppression of nucleation by discrete size polydispersity in colloids and bulk metallic glass-forming liquids is explored. We seek to elucidate the initial stages of crystal nucleation in supercooled liquids and colloids above their glass transition, and to relate nucleation phenomena to local structural and dynamical features in these systems. We further seek to understand the role of discrete size polydispersity in suppressing nucleation, and to compare this phenomenon in colloids and a model bulk metallic glass-forming liquid. This project addresses fundamental, ground-breaking issues in the solidification of liquids, particularly bulk metallic glass-forming liquids, and colloidal suspensions. The results of this project are critical to understanding and controlling the properties and behavior of technologically-relevant materials, in particular for space applications. Our work will provide strategies for designing colloidal materials with specific interactions to cause their solidifcation via self-assembly into unique crystal structures for target applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New materials designed for space exploration and long-term space travel have strategic requirements related to strength, weight, durability, cost, etc. The science of solidification provides the necessary understanding for controlling the properties and behavior of technologically-important materials and making new materials for space exploration.
Task Progress:	In this project, we seek to elucidate the initial stages of crystal nucleation in supercooled liquids and colloids above their glass transition, and to relate nucleation phenomena to local structural and dynamical features in these systems. We further seek to understand the role of discrete size polydispersity in suppressing nucleation, and to compare this phenomenon in colloids and a model bulk metallic glass-forming liquid. As our model liquid, we chose the Dzugutov pairwise interaction potential, which is constructed to suppress formation of cubic crystal symmetries and thereby promote stability of the supercooled liquid. Our goals in this initial year of the grant focused on carrying out initial simulations of the Dzugutov liquid to characterize the spatially heterogeneous nature of the dynamics and relate it to local structure, and develop and test the analysis codes needed to quantify local order for studies of nucleation. Experimentally, our goals included establishing the experimental tools and parameters for carrying out confocal laser scanning microscopy experiments and conducting initial shear experiments. Our accomplishments in this first year are briefly summarized below. A) Extensive molecular dynamics simulations of a supercooled liquid modeled by the Dzugutov potential are underway. We are completing a detailed investigation of the local particle ordering in which we have characterized the degree of icosahedral ordering in the liquid as a function of temperature via cluster analysis. We have analyzed aspects of spatially heterogeneous dynamics and showed how cooperatively rearranging particles moving in one-dimensional “strings” move along the surfaces of, and between, clusters of icosahedrally-ordered particles. We performed an analysis of dynamic facilitation and showed that particle motion in any given region of the sample is dynamically facilitated, that is, facilitated by motion of particles in neighboring regions. This finding supports recent theoretical predictions by Garrahan and Chandler. We have carried out a detailed test of the Adam-Gibbs theory of structural relaxation, and tested our hypothesis that the average string length is an appropriate measure of the size of a cooperative domain postulated by Adam and Gibbs. We have developed analysis tools for quantifying higher order local bond orientation parameters to determine the crystal structure of nucleating clusters in simulations and the experiments described below. B) Confocal laser scanning microscopy (CLSM) studies of the crystallization of micron sized colloidal silica and poly(methyl methacrylate) (PMMA) suspensions were initiated. Density and refractive index matched solvent mixtures were developed for both the silica and PMMA systems. At low volume fractions, strong attractive interparticle interactions lead to the observation of glass-like dynamics in the silica colloidal system. Image processing methods for the location of colloidal particles within a CLSM image volume have been applied to regions in which crystallization was observed. Qualitative differences between crystallization in the silica and PMMA colloidal systems have been observed. We have observed three-dimensional ordering of suspensions under quiescent conditions and as they are subjected to oscillatory shear flow. Qualitatively, we find that shear flow induces nucleation and growth of colloidal crystals in the initially amorphous suspensions. The pair correlation function and local bond orientation parameters have been computed from coordinate files generated by image processing of the acquired CLSM image volumes. The strain rate and amplitude dependence of these measures of local ordering have been quantified for the silica colloidal system. Our recent experiments have used direct visualization via confocal microscopy to study the effect of shear and gravitational fields in directing the crystallization of PMMA colloids suspended in nonpolar organic solvents. In shear flow experiments, we have designed a parallel plate flow device that is integrated into the confocal set-up and provides the ability to observe the sheared structures in situ. The effect of varying the frequency and amplitude of an oscillatory field has been probed. Quantitative image processing methods have been applied to the images collected. We extracted measures of local ordering such as the pair correlation function and local bond orientational parameters. These measures quantify the effect of nonuniformities in the crystallization. A method to examine stacking faults is in development. The shear flow results described above are being prepared for publication. Colloidal crystals formed from sedimented colloids have been made to assess the effect of a gravitational field on crystal structure. By using solvent mixtures and by varying the particle size, we control the sedimentation rate of the colloids and thus the strength of the gravitational field. We have observed vacancies, stacking faults, and heterogeneities in the degree of crystallization of these sedimented colloidal crystals. We have begun applying quantitative image studies to these results. Several publications have been submitted or published during this grant period, and many presentation have been given (see attached bibliography).
Bibliography Type:	Description: (Last Updated: 04/09/2008)

Abstracts for Journals and Proceedings	Y. Gebremichael, M. Vogel, M. Bergroth, F.W. Starr and S.C. Glotzer "Applications of Inherent Structure Formalism and Relation Between Spatially Heterogeneous Dynamics and the Adam-Gibbs Theory" manuscript submitted for publication to J. Physical Chemistry no index , Jul-2004
Abstracts for Journals and Proceedings	Bergroth, M.; Vogel, M.; Glotzer, S.C. "Examination of dynamic facilitation in model supercooled liquids" submitted for publication in J. Physical Chemistry no index , Oct-2004
Articles in Peer-reviewed Journals	Gebremichael, Y.; Vogel, M.; Glotzer, SC "Particle dynamics and the development of string-like motion in a mono-atomic, supercooled liquid" J. Chemical Physics, 120 (9): 4415-4427 (2004) , Jan-2004
Dissertations and Theses	Gebremichael, Y. "Spatially heterogeneous dynamics and string-like correlated motion in supercooled liquids and polymer" PhD Dissertation Mar-2004
Presentation	Glotzer, S.C. "Cages, Clusters and Conga Lines: Liquids Behaving Badly" Department of Chemistry Colloquium, University of Wisconsin Jan-2004
Presentation	Bergroth, M.; Keys, A.S.; Vogel, M.; and Glotzer, S.C. "Dynamical Heterogeneity, Dynamic Facilitation, and Frustration-Limited Domains in a Simulated Glass-Forming Liqui" March 2004 Meeting of the American Physical Society, Montreal Mar-2004
Presentation	Gebremichael,Y.; Vogel, M.; and Glotzer, S.C. "Particle Dynamics and the Development of String-like Motion in a Dzugutov Liquid" March 2004 Meeting of the American Physical Society, Montreal Mar-2004
Presentation	Glotzer, S.C. "Spatially Heterogeneous Dynamics in Strong and Fragile Liquids" Invited talk at Workshop on Unifying Concepts in Glass Physics, Bangalore, India Jun-2004
Presentation	Glotzer, S.C. "Tutorial on Spatially Heterogeneous Dynamics" Invited Lecture in Short Course on the Glass Transition, Unifying Concepts in Glass Physics, Bangalore, India Jun-2004
Presentation	Glotzer, S.C. "Beyond Squish: The New Science of Colloids" Invited Talk at National Academy of Science Beckman Frontiers of Science Symposium, Irvine, CA Nov-2004
Presentation	Keys, A.S.; and Glotzer, S.C. "Effect of Local Icosahedral Ordering on Supercooled Liquid Stability Against Nucleatio" AICHE Annual Fall Meeting, Austin, TX Nov-2004
Presentation	Bergroth, M.; Keys, A.S.; Gebremichael, Y.; Vogel, M.l and Glotzer, S.C. "Spatially Heterogeneous Dynamics, Dynamic Facilitation, and Frustration-Limited Domains in a Simulated One-Component Supercooled Liquid Near the Glass Transition" AICHE Annual Fall Meeting, Austin, TX Nov-2004
Presentation	Solomon, T.; and Solomon, M.J. "Shear-directed assembly of colloidal crystals visualized by confocal microscopy," AIChE National Meeting" AIChE National Meeting, Austin TX Nov-2004
Presentation	Solomon, M.J. "Flow-assisted assembly of ordered colloidal structures" Department of Chemical Engineering Colloquium, Michigan Tech University Oct-2004